Iteration 4:
RAG Coaching Agent For At-Home Rehabilitation Exercise Utilizing Human Pose Estimation

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(Dated: March 24, 2026)

I. PROJECT PURPOSE

At-home rehabilitation is a critical yet under-served phase of injury recovery. Patients discharged from clinical care are typically given a printed or verbal exercise regimen and expected to execute it correctly, consistently, and safely but without the supervision of a physical therapist / medical professional. This project addresses that gap by building a multimodal AI coaching agent that combines computer-vision-based movement analysis with a retrieval-augmented generation (RAG) framework to deliver real-time, clinically grounded feedback during at-home physiotherapy exercises.

The system integrates three coupled subsystems. First, a Temporal Convolutional Network (TCN) trained on the Qualcomm Exercise Video Dataset (QEVD) classifies movement quality from 3D pose estimates extracted via MediaPipe, identifying exercise identity, form errors, speed, and range of motion across 148 exercise categories. Second, a RAG pipeline grounded in curated physiotherapy literature including NICE Clinical Guidelines, StatPearls, and the American Academy of Orthopaedic Surgeons, retrieves evidence-based context to support patient feedback generation. Third, the integration layer bridges the gap and applies temporal aggregation, severity classification, and three-tier routing logic to yield appropriate coaching cues.

Our methodology balances a core trade-off: First, generating medically grounded patient feedback from rehabilitation literature, for which the pipeline requires a 10 second latency that dis-allows back and forth LLM conversation; second, also remaining deployable on consumer hardware, especially inexpensive smart phones, and third operating with full automation without any need for a live medical professional. We implement a functional end-to-end system prototype that is accessible to the every day patient and demonstrate its utility in several live demo use cases. If we are able to take advantage of greater computational resources as well as additional time to extend the project, we could fine-tune both the computer vision pipeline as well as the RAG and integration layers to fill out the user interface as well as the patient experience and to optimize the overall system performance across diverse exercises and rehabilitation regimes.

II. PROBLEM STATEMENT

A. Defining the Problem

Injury rehabilitation fundamentally depends on the correct, consistent execution of prescribed exercises over an extended recovery period. In clinical settings, a licensed physiotherapist provides real-time observational feedback such as identifying compensatory movements, unsafe loading patterns, and deviations from prescribed form, and then adjusts the exercise program as the patient progresses. Once discharged to home-based care, patients lose access to this supervisory feedback loop entirely.

This project frames the core challenge as a **multimodal, multi-task machine learning problem** combining three distinct components:

- **Multi-label movement quality classification (computer vision).** Given a variable-length video sequence of a patient performing a rehabilitation exercise, simultaneously classify exercise identity (148-class), form errors, movement speed, range of motion, torso orientation, and lateral direction.
- **Retrieval-augmented coaching feedback generation (LLM).** Given a detected movement error and its clinical context, retrieve the most relevant physiotherapy guidance from a curated medical corpus and generate patient-facing coaching feedback that is accurate, safe, and actionable. This is a domain-specific, knowledge-grounded text generation task with hard safety constraints.
- **Stateful multi-agent orchestration (integration layer).** Route detected errors through a three-tier response system: cached responses for known patterns, RAG-grounded LLM generation for novel errors, and full multi-step agent reasoning for persistent or complex cases—while applying temporal aggregation and cooldown logic to suppress noise and prevent feedback fatigue.

B. Significance

Clinical motivation. Incorrect exercise execution during home rehabilitation carries two distinct risks. The first is physical: compensatory movements and improper

loading can aggravate the original injury, introduce secondary injuries, or lead to maladaptive motor patterns that delay recovery. The second is behavioral: patients who are unsure whether they are performing exercises correctly are more likely to discontinue their regimen. Patient compliance or adherence to the full exercise frequency and duration prescribed by a clinician, is consistently identified as one of the primary determinants of rehabilitation outcomes, and one of the most persistent barriers to successful at-home recovery [1–3].

Research motivation. The intersection of human pose estimation, multi-task deep learning, and RAG-grounded clinical NLP seems underexplored as an integrated system. Prior work in AI-assisted exercise coaching has largely focused on narrow domains such as yoga posture correction [4**? ? ?** , 5] or single-task form classification, without coupling movement analysis to a clinically grounded feedback generation pipeline. Recent evaluations of RAG systems in medicine [**?**] highlight both the promise and the risks of LLM-generated clinical guidance, underscoring the importance of rigorous retrieval design.

Practical impact. A system that can act as a virtual physiotherapy assistant (VPA), observing exercise performance, identifying errors, and delivering evidence-based corrective feedback in real time, has the potential to extend the reach of physical therapy beyond clinical hours and clinic walls. By tuning the agent to a patient's specific injury type and their clinician's prescribed protocol, the system can provide individualized guidance that approximates the role of a supervising professional, improving both the safety and the efficacy of at-home rehabilitation at scale.

III. DATA COLLECTION AND PREPARATION

A. Data Sources

The project draws on three independent data sources corresponding to its core subsystems: a computer-vision (CV) movement-quality classifier, a Retrieval-Augmented Generation (RAG) recommendation engine, and the integration layer that routes between them.

a. RAG corpus — primary PT guidelines. Physiotherapy guidelines and rehabilitation protocols were manually curated from 22 authoritative medical websites and clinical sources, spanning five anatomical regions: knee, hip, shoulder, low back, and mid-back. Sources include NICE Clinical Guidelines, StatPearls (NCBI Bookshelf), the MOON Knee Group, Mass General Brigham, NHS Trust Physiotherapy (MSK Patient Portals), the American Academy of Orthopaedic Surgeons OrthoInfo library, Physiopedia, the University of Delaware Physical Therapy Clinic, VA/DoD Clinical Practice Guidelines, Allina Health Orthopedics, ChoosePT, E3 Rehab, the PreHab Guys, Squat University, and the Movement as Medicine Programme, among others. The primary corpus consists of **65 HTML pages** and **210 PDF documents** collected from these medical-approved websites.

b. RAG corpus — additional PT guidelines. A supplementary corpus was constructed from peer-reviewed textbooks and Physiopedia reference articles to broaden the clinical knowledge base. This comprises **6 HTML pages** and **29 PDF documents**, including titles such as *Therapeutic Exercise: Foundations and Techniques* (Kisner & Colby), *Therapeutic Exercise for Musculoskeletal Injuries* (Houglum), *Daniels and Worthingham's Muscle Testing* (Hislop et al.), *Home Exercise Programs for Musculoskeletal and Sports Injuries* (Wender & Wyss), NICE/JOSPT clinical practice guidelines for neck pain and whiplash, and Physiopedia reference pages for hip, shoulder, and tendinopathy rehabilitation. Combined, the full RAG corpus contains **71 HTML pages** and **239 PDF documents** across both sub-corpora.

c. Pose video dataset — QEVD. The computer-vision pipeline was trained on the **Qualcomm Exercise Video Dataset (QEVD)** [6], a large-scale benchmark introduced at NeurIPS 2024 for live fitness coaching research. QEVD contains over **474 hours of video** covering **148 exercises** and their fine-grained variations, including varied pacing, common mistakes, and modified form. Short-clip videos (approximately 5 seconds each) are annotated with 1M+ question-answer pairs and 650k+ corrective feedback labels; long-range session videos (>3 minutes) are annotated with live interactive feedback. Quantitative labels for speed (reps per second) and range of motion are available for a subset of exercises.

The QEVD event-log exports used for model development consist of **2,802 compressed event files** in total: 538 files from the test partition (217 containing valid event frames) and 2,264 files from the validation partition (1,292 containing valid event frames), for a combined **1,509 clips with annotated exercise events**.

d. Synthetic data — integration layer testing. Two categories of synthetic data were generated to support development and testing of the integration layer:

- **Synthetic continuous frame sequences.** Because QEVD event logs provide only single-frame snapshots of detected mistakes, synthetic 120-frame video sequences were programmatically generated from the event logs to produce the continuous frame streams required by the integration layer's temporal aggregation logic. In total, **1,509 synthetic videos** were generated (217 from the test partition, 1,292 from the validation partition).

- **Synthetic ground-truth coaching cues.** A ground-truth annotation file (`ground_truth_coaching_cues.json`) was developed by synthetically generating representative mistake labels, exercise identities, and associated coaching cues to establish a reference for integration layer evaluation and threshold tuning.

B. Data Description

a. RAG corpus. Document count and length vary by anatomical region; the shoulder and knee are the most densely represented across both sub-corpora, while the mid-back and neck have narrower coverage. The primary PT guidelines contain clinical handouts and web-based patient education materials (typically 500–3,000 words per document), while the additional PT guidelines comprise substantially longer textbook chapters (up to several hundred pages per title after text extraction). A curated set of **28 validated test questions** was constructed to cover six clinically relevant query categories: exercise instructions, safety and pain monitoring, expected outcomes, exercise progression, contraindications, and return-to-sport criteria. Each question carries two to three expert-authored ground-truth answers derived from the source literature.

b. Pose dataset. Each QEVD clip is processed into a compressed `.npz` archive containing a landmark array of shape $(T \times 33 \times 3)$ (x, y, z coordinates) and a visibility array of shape $(T \times 33)$, where T is the number of retained frames after temporal trimming. After feature engineering (Section III C), each frame is represented as a **198-dimensional vector**. Labels include a single-class exercise identity label drawn from 148 exercise categories, a multi-hot mistake vector, and ordinal labels for speed, range-of-motion bin, height, torso rotation, and direction.

c. Synthetic integration layer dataset. Each of the 1,509 synthetic videos consists of **120 frames** simulating a realistic workout clip. Frames 0–14 represent good form with no detected mistakes; the target mistake(s) from the source event log begin appearing from frame 15 onward with a 70% per-frame detection rate and ± 0.1 confidence variation to mimic natural detector noise. Transient minor mistakes are injected at a 10% rate across all frames to simulate real-world false-positive noise. This design ensures that the persistent-mistake criterion (appearing in $\geq$30% of frames for $\geq$3 seconds) is reliably triggerable, enabling systematic threshold tuning.

C. Preprocessing Steps

1. RAG Pipeline

a. Primary PT guidelines. HTML files were parsed and cleaned using `BeautifulSoup4` to remove navigation elements, headers, and non-content markup. PDF documents were parsed using `pypdf`, with whitespace normalisation and removal of PDF extraction artefacts applied post-extraction. Each cleaned document was segmented into semantically coherent chunks, embedded using **Sentence Transformers**, and stored in a **ChromaDB** vector database. At query time, the top-k most similar chunks are retrieved via cosine similarity and injected into a structured slot-based prompt template before generation. No stemming, lemmatisation, or stopword removal was applied, as the domain-specific clinical vocabulary must be preserved for accurate retrieval.

b. Additional PT guidelines. Textbook PDFs and MHTML pages were converted to plain-text using `pypdf` and custom extraction scripts. The cleaned text files (stored under `data/additional_pt_guidelines/cleaned/`) range from 106 KB (clinical guidelines) to 4.49 MB (full textbook transcriptions) and were preprocessed with the same whitespace normalisation pipeline as the primary corpus before ingestion into ChromaDB.

c. RAG retrieval strategy comparison. To quantify the value of the retrieval design, an early experiment compared *traditional RAG* (vector embeddings + similarity search, top-3 chunks, avg. latency $\approx$7 s) against an *agent-based* approach in which a Claude-powered agent iteratively traversed the document filesystem until it judged the gathered context sufficient (avg. 6.2 iterations, avg. latency $\approx$28 s, avg. 15,907 tokens per query). Both approaches achieved 100% source-retrieval accuracy on a six-query pilot set, but the agent-based approach consumed roughly 20$\times$ more tokens per query at 67$\times$ higher cost with no commensurate gain in answer quality. Traditional RAG was therefore adopted as the production retrieval strategy.

2. Computer Vision Pipeline

Each QEVD video clip undergoes the following sequential preprocessing steps.

a. Frame-level pose extraction. MediaPipe's **Pose-Landmarker** ("full" variant, float16 weights) is applied to every frame in IMAGE running mode, yielding normalised screen-space (x, y) coordinates, a depth estimate z, and a per-landmark visibility score for all 33 body keypoints. Extraction is parallelised across clips using a `ProcessPoolExecutor` with spawn-based workers to avoid OpenMP thread contention.

b. Temporal stability trimming. A sliding-window algorithm scans the opening two seconds of each clip to identify the earliest 10-frame window satisfying:

- $\geq$65% of keypoints with visibility $\geq$ 0.5 (good-fraction threshold),
- median inter-frame jitter $\leq$ 0.035 in normalised units.

If such a window is found, all frames before it are discarded. Clips failing this criterion are either skipped or retained in full as a fallback. This step eliminates unstable camera-setup frames that would corrupt temporal features.

c. Pose normalisation and feature engineering. Each retained frame is normalised and transformed into a 198-dimensional feature vector:

1. **Translation:** subtract the hip-centre coordinate (mean of landmarks 23 and 24) from all (x, y) pairs.
2. **Scale:** divide by the Euclidean distance between the left and right shoulders (landmarks 11 and 12), clipped to a minimum of 10^{-3}.
3. **Feature concatenation:** normalised xy positions ($33 \times 2 = 66$), frame-to-frame velocity in xy ($33 \times 2 = 66$), depth z (33), and visibility (33), yielding a 198-dimensional vector per frame.

d. Memory-mapped storage. Processed feature arrays are stored as float16 memory-mapped files (`X.f16.mmap`) alongside metadata arrays, enabling large-batch DataLoader access without loading the full dataset into RAM.

e. Train/validation/test splits. Data are stratified by exercise identity. For each exercise class with ≥8 clips, 2 clips are reserved as a held-out test set. Of the remaining clips, 8% form the validation set; the rest are used for training. A fixed random seed ensures reproducibility.

IV. SELECTION OF MACHINE LEARNING AND LLM MODELS

A. Model Considerations

The system requires three model-level decisions: a temporal sequence classifier for movement quality analysis, a large language model (LLM) for RAG-grounded coaching feedback generation, and a routing and aggregation strategy for the integration layer that bridges real-time CV output to LLM feedback.

1. Movement Quality Classifier

Because the pose feature representation is a variable-length time series, the model family must capture temporal dependencies across frames. The following architectures were considered:

- **Recurrent networks (LSTM/GRU):** strong at sequential modelling but difficult to parallelise during training and prone to vanishing gradients over long sequences.
- **Transformer-based sequence models:** expressive but computationally expensive and prone to overfitting on a dataset of ∼1,500 labelled clips.
- **Temporal Convolutional Network (TCN):** fully convolutional, efficiently parallelisable, and capable of capturing multi-scale temporal patterns via dilated causal convolutions. TCNs offer competitive performance to LSTMs on action-recognition benchmarks with faster training on CPU/MPS hardware.

A TCN was selected as the best balance of modelling capacity, training efficiency, and deployment suitability on consumer hardware (Apple Silicon MPS or CPU).

2. LLM for RAG Generation

Three language models were evaluated for RAG-grounded coaching feedback generation (full results in Section VI):

- **Claude Sonnet 4.5** (Anthropic API) — a high-capability commercial model; included as a quality upper-bound.
- **Gemma3:4b** (Google, served locally via Ollama) — a compact open-weights model for zero-cost local deployment.
- **Qwen2.5:7b** (Alibaba, served locally via Ollama) — a larger open-weights model to assess whether additional parameters improve quality relative to Gemma3.

Local models eliminate per-query API costs and keep patient data on-premise, which is important for a healthcare application. Their inclusion was motivated by both cost and privacy considerations. **Claude Sonnet 4.5** was ultimately selected as the primary generation model based on its superior response completeness and clinical accuracy, as detailed in Section VI.

3. Integration Layer Routing Strategy

The integration layer mediates between the CV pipeline and the LLM system, performing temporal aggregation, deduplication, and tier routing. Rather than forwarding every detected mistake immediately to the LLM, it applies a three-stage filtering pipeline to suppress noise and select the appropriate response tier:

1. **Temporal aggregation.** A 10-second sliding window (150 frames at 15 FPS; minimum 75 frames to trigger analysis) tracks each detected mistake type. A mistake qualifies as *persistent* if it appears in ≥25% of frames, with average confidence ≥0.30, and persists for ≥2.5 seconds.
2. **Deduplication and cooldown.** Once a mistake type has been addressed, a 10-second inter-coaching cooldown prevents redundant feedback. Re-coaching is triggered only if the same mistake persists for ≥20 seconds after initial coaching.
3. **Severity classification and tier routing.** Each qualifying mistake is classified as **high**, **medium**, or **low** severity based on keyword matching and the CV quality score. The routing decision follows a three-tier hierarchy:

- **Tier 1 (Cache, ≈50 ms):** Common, pre-identified mistake patterns served from a disk-backed response cache (`diskcache`).
- **Tier 2 (RAG + LLM, ≈1–2 s):** High or medium severity mistakes without a cached response; RAG retrieves context and the LLM generates a personalised coaching cue.
- **Tier 3 (Full agent reasoning, ≈3–5 s):** Re-coaching situations or complex patterns requiring multi-step LangGraph reasoning.

B. Final Model Selection

1. *Movement Quality Classifier: PoseTCNTyped*

The **PoseTCNTyped** architecture was finalized for the CV pipeline. It consists of six stacked TCN blocks with hidden dimension 256 and dilation factors $d \in \{1, 2, 4, 8, 16, 32\}$, followed by global average pooling and eight task-specific output heads:

- **Exercise identity head** (`CrossEntropyLoss`) — single-label classification across 148 exercise categories.
- **Mistake type head** (`BCEWithLogitsLoss`, multi-label) — detects one or more simultaneous form errors (e.g. knee valgus, incomplete arm raise, lumbar hyperextension).
- **Speed, ROM, height, torso, direction heads** (`CrossEntropyLoss`, ordinal or 3-class) — encode quantitative movement attributes used downstream for personalised coaching.
- **No-issue head** (`BCEWithLogitsLoss`, binary) — predicts whether the clip contains no observable error.

The total training loss is a weighted sum:

$$\begin{aligned}\mathcal{L} = 1.0\,\mathcal{L}_{\text{ex}} + 0.8\,\mathcal{L}_{\text{mist}} + 0.3\,(\mathcal{L}_{\text{speed}} + \mathcal{L}_{\text{rom}}) \\ + 0.2\,(\mathcal{L}_{\text{height}} + \mathcal{L}_{\text{torso}} + \mathcal{L}_{\text{dir}} + \mathcal{L}_{\text{no-issue}})\end{aligned} \tag{1}$$

The multi-task design provides richer supervisory signal and acts as an implicit regulariser. Dropout ($p = 0.2$) and gradient clipping (max norm = 1.0) are applied during training.

2. *LLM: Claude Sonnet 4.5*

Based on the comparative evaluation described in Section VI, **Claude Sonnet 4.5** was selected as the primary generation model for its leading performance on response completeness (46.9%), fact coverage (41.5%), and semantic coverage (69.7%) across the 28-question physiotherapy evaluation set.

V. MODEL DEVELOPMENT AND TRAINING

A. Architecture and Configuration

1. *Computer Vision Pipeline*

The movement quality classifier is a multi-task Temporal Convolutional Network (TCN) operating on 198-dimensional pose feature vectors. Each vector encodes, per frame, the normalised (x, y) positions of 33 body keypoints (66 features), frame-to-frame velocity in xy (66 features), per-landmark depth estimate z (33 features), and per-landmark visibility score (33 features).

The PoseTCNTyped architecture consists of six stacked dilated causal TCN blocks with a hidden dimension of 256 and exponentially increasing dilation factors $d \in \{1, 2, 4, 8, 16, 32\}$, yielding a theoretical receptive field of $2(2^6 - 1) + 1 = 127$ frames. Each block applies a 1-D convolution along the temporal axis with causal padding to prevent information leakage from future frames. Dropout ($p = 0.2$) is applied within every TCN block as the primary regularization mechanism. Following the six TCN blocks, global average pooling collapses the temporal dimension into a single feature vector, which is then fed to eight independent task-specific output heads:

- **Exercise identity** (CrossEntropyLoss) — 148-class single-label classification.
- **Mistake type** (BCEWithLogitsLoss, multi-label) — simultaneous detection of one or more form errors (e.g. knee valgus, lumbar hyperextension).
- **Speed, range of motion, height, torso** (CrossEntropyLoss, 5-class ordinal each) — quantitative movement attributes used downstream for personalised coaching.
- **Direction** (CrossEntropyLoss, 3-class) — neutral, clockwise, or counter-clockwise.
- **No-issue** (BCEWithLogitsLoss, binary) — predicts whether the clip contains no observable error.

The joint training loss is a task-weighted sum that prioritizes exercise identity and mistake detection.

2. *LLM: Claude Sonnet 4.5 with RAG Grounding*

Claude Sonnet 4.5 is deployed as a pre-trained, frozen model accessed via the Anthropic API; no fine-tuning was performed. Rather than adapting the model weights to the physiotherapy domain, clinical knowledge is injected via the RAG pipeline.

B. Training Process

1. Dataset Splits

All data splits for the CV classifier are stratified by exercise identity to ensure every class is represented across partitions. For each exercise class containing at least eight clips, exactly two clips are reserved as a held-out test set. Of the remaining clips, 8% are assigned to the validation set and the remainder constitute the training set. A fixed random seed is used throughout to ensure reproducibility. The resulting splits draw from 1,509 clips across 217 test-partition and 1,292 validation-partition event files derived from QEVD. The LLM component is evaluated on a fixed set of 28 expert-authored physiotherapy test questions with two to three ground-truth answers each; this set is held out entirely from retrieval index construction and used solely for final evaluation.

2. Overfitting Mitigation

The primary over-fitting risk in the CV pipeline arises from the relatively small number of labelled clips (∼1,500) relative to model capacity. Three complementary strategies are employed to address this. First, per-block dropout ($p = 0.2$) within the TCN stochastically zeros activations during training, preventing co-adaptation of filters across blocks. Second, the multi-task objective itself should act as a structural regularizer: by simultaneously supervising eight output heads, the shared TCN backbone is constrained to learn representations that generalize across movement attributes rather than overfitting to the dominant exercise-identity signal alone. Third, gradient clipping (max norm = 1.0) prevents individual large-gradient updates from destabilizing the shared feature extractor.

VI. EVALUATION AND COMPARISON

This section reports the evaluation results for the two primary model decisions: the RAG-grounded LLM component and the integration layer routing behaviour. Because the project is oriented toward a consumer-facing rehabilitation application rather than a standalone research benchmark, the evaluation emphasises practical criteria — clinical accuracy, response completeness, and real-time feasibility — alongside standard automated metrics.

A. LLM Evaluation

1. Metrics

Three categories of metrics were used to evaluate the LLMs against the 28 validated physiotherapy test questions:

a. Quality metrics measure how well a response covers the clinically correct information:

- **Fact coverage** — fraction of expert ground-truth factual claims present in the model response.
- **Semantic coverage** — embedding-based similarity between the response and ground-truth answers, rewarding breadth of coverage.
- **Completeness** — composite score combining fact coverage (70%), semantic coverage (20%), and a comprehensiveness bonus for well-structured, actionable responses (10%).

b. Safety metrics are binary pass/fail criteria critical for a healthcare context:

- **Safety pass rate** — whether the response avoids contraindicated or clinically unsafe advice.
- **Professional referral rate** — whether the response appropriately recommends consulting a healthcare professional.
- **Medical accuracy** — binary pass/fail on clinically verifiable factual claims.

c. Operational metrics reflect deployment constraints:

- **Latency** — average wall-clock response time (seconds), and P95 latency.
- **Cost** — total USD per 28-question evaluation run.

2. Results

Table I summarises the final evaluation results across all three models.

All three models achieved a **100% safety pass rate**, **100% professional referral rate**, and **100% medical accuracy** on the test set, confirming that RAG grounding reliably suppresses unsafe or unsupported clinical claims regardless of the underlying model.

Claude Sonnet 4.5 led on every quality metric, achieving the highest overall score (58.93) through substantially stronger fact coverage (41.5% vs. ≈30–31% for local models) and semantic coverage (69.7% vs. ≈48–53%). Its average latency (8.3 s) was also faster than both local models (12–14 s), despite routing through an external API. The main trade-off is per-query cost; at $0.244 per

TABLE I. LLM evaluation on 28 validated physiotherapy questions. Best values in **bold**.

Model	Overall	Fact Cov.	Sem. Cov.	Completeness	Latency (s)	Cost ($)
Claude Sonnet 4.5	**58.93**	**0.415**	**0.697**	**0.469**	**8.29**	0.244
Qwen2.5:7b	54.18	0.314	0.531	0.357	13.54	**0.000**
Gemma3:4b	54.04	0.298	0.476	0.341	12.32	**0.000**

28-question run, this is acceptable for a healthcare application where response quality has a direct impact on patient safety and recovery outcomes.

Qwen2.5:7b narrowly outperformed Gemma3:4b on both fact and semantic coverage, suggesting that model scale provides modest gains within the open-weight tier. However, neither local model approached Claude Sonnet 4.5 on completeness. Both remain viable for a privacy-first, fully offline deployment scenario where API access is unavailable.

B. CV Classifier Evaluation

The primary evaluation metric for the **PoseTCN-Typed** movement quality classifier is held-out **exercise identity accuracy**, which is reported at the end of each training epoch on the stratified test split (2 held-out clips per exercise class). Secondary metrics include per-task training and validation loss for each of the eight output heads (exercise, mistake type, speed, range of motion, height, torso, direction, and no-issue), allowing individual task performance to be monitored independently. Because the project is application- focused, the most operationally relevant downstream metric is whether the classifier correctly identifies the exercise type and flags meaningful form errors — both of which feed directly into the integration layer's routing decisions.

C. Integration Layer Evaluation

The integration layer was validated on **1,509 synthetic QEVD-derived videos** (217 test, 1,292 validation) using the following system-level metrics:

- **Coaching event rate** — target 0.27–0.32 events per video; observed values fell within this range, indicating that the temporal aggregation thresholds are appropriately calibrated to suppress transient noise without under-coaching.
- **Video coverage** — approximately 27–32% of videos received at least one coaching event, consistent with the expected proportion of clips containing persistent form errors.
- **Tier distribution** — without a pre-populated Tier 1 cache, 44–87% of events were routed to Tier 2 (RAG + LLM) and 12–56% to Tier 3 (full agent reasoning), confirming the routing logic behaves as designed.
- **Severity distribution** — 6–41% high-severity, 26–50% medium-severity, and 33–44% low-severity events across test runs, reflecting the diversity of error types present in the QEVD dataset.

These results validate that the integration layer correctly bridges real-time CV output to the appropriate response tier without generating excessive or redundant feedback — a core requirement for a usable consumer rehabilitation application.

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